

INTERNSHIP REPORT
ON
**DATA ANALYTICS & ARTIFICIAL INTELLIGENCE
ESSENTIALS**

Through Cisco Networking Academy

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement for the degree of
Bachelor of Technology in Computer Science and Engineering

Submitted By

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Batch: 3CSE2

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CANDIDATE'S DECLARATION

I, Gopal Agarwal, hereby declare that the report entitled “Data Analytics & Artificial Intelligence Essentials through Cisco Networking Academy” has been prepared and completed by me in fulfilment of the academic requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering. This report presents my learning, understanding, and experience gained through three programs completed under the Cisco Networking Academy initiative: Data Analytics Essentials, Introduction to Modern AI, and Apply AI: Analyze Customer Reviews. I have made sincere efforts to present the information accurately and systematically. All sources and references used in preparing this report have been appropriately acknowledged. I further declare that this report is my original academic work and has not been submitted to any other institution for the award of any degree or diploma. I shall be responsible for any copyright or academic-integrity violation arising from the contents of this report.

Signature: _____

Student Name: Gopal Agarwal

Roll No.: 2410030241

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me during the completion of the Data Analytics Essentials, Introduction to Modern AI, and Apply AI: Analyze Customer Reviews programs offered through the Cisco Networking Academy. This learning experience provided me with an opportunity to develop a strong understanding of data analytics and artificial intelligence and their growing importance in the modern technology-driven environment.

I am thankful to IILM University and the Cisco Networking Academy program for providing this valuable learning opportunity. I am also grateful to Ms. Swati Vashisht, Instructor, IILM University, for her guidance and academic support during the Data Analytics Essentials program, and to the Cisco Networking Academy for facilitating the Introduction to Modern AI and Apply AI: Analyze Customer Reviews courses. The knowledge gained through these programs has helped me understand the fundamentals of working with data and artificial intelligence, and has strengthened my interest in analytics and technology.

I would also like to thank my faculty members, classmates, friends, and family for their continuous encouragement and support. Their motivation helped me remain focused throughout the learning process and successfully complete all three programs.

1. INTERNSHIP / PROGRAM COMPLETION CERTIFICATES

The certificates issued for the successful completion of the three programs undertaken — Data Analytics Essentials, Introduction to Modern AI, and Apply AI: Analyze Customer Reviews — are attached below as documentary proof.

1.1 Data Analytics Essentials

Offered by IILM University through the Cisco Networking Academy program. Instructor: Ms. Swati Vashisht. Completion Date: 14 June 2026. Certificate ID: 3c0dff58-375b-4273-aeb4-0d81627c82ae.

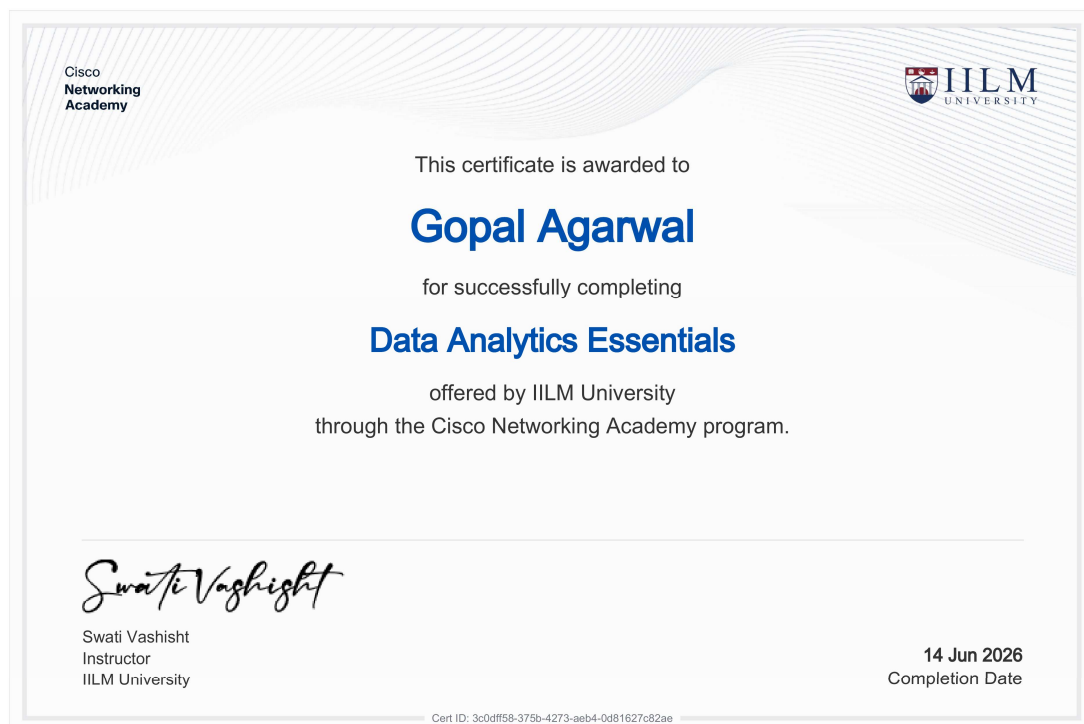


Figure 1.1 – Certificate of Completion: Data Analytics Essentials

1.2 Introduction to Modern AI

Offered by Networking Academy through the Cisco Networking Academy program. Director: Lynn Bloomer.
Completion Date: 12 June 2026. Certificate ID: daf617bb-4ffa-4482-8b2a-6c17fa384ff3.

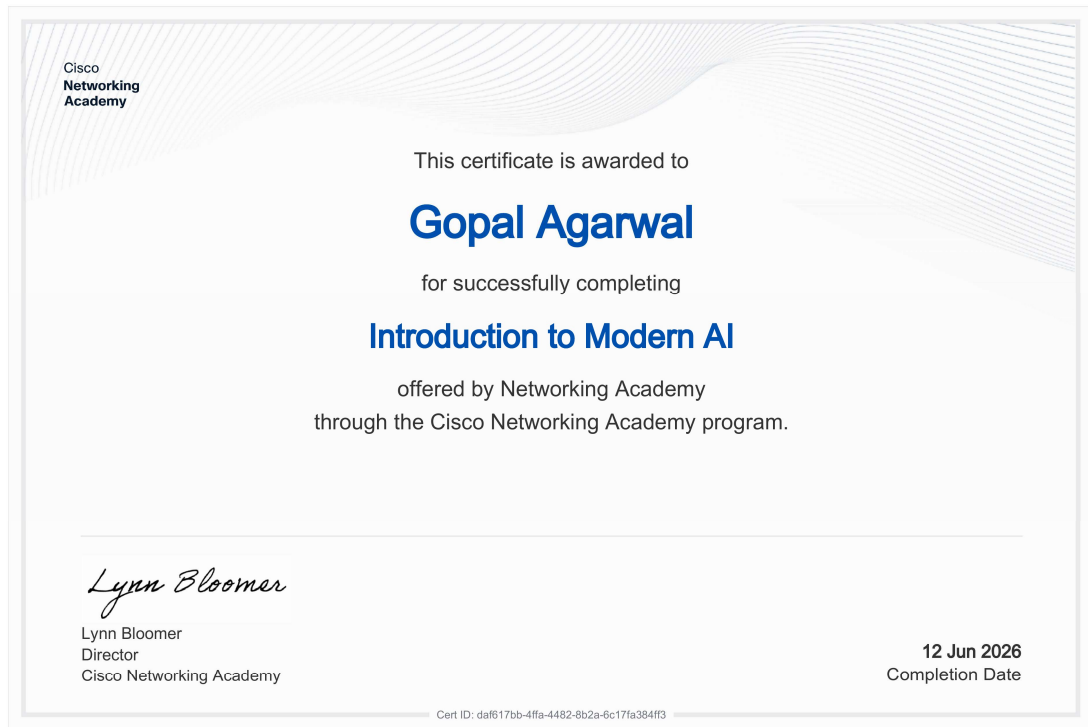


Figure 1.2 – Certificate of Completion: Introduction to Modern AI

1.3 Apply AI: Analyze Customer Reviews

Offered by Networking Academy through the Cisco Networking Academy program. Director: Lynn Bloomer.
Completion Date: 12 June 2026. Certificate ID: 7e59bd23-d10a-4d85-a2b6-238c3569a919.



Figure 1.3 – Certificate of Completion: Apply AI: Analyze Customer Reviews

2. PROJECT / TRAINING DESCRIPTION

2.1 Introduction

Data and artificial intelligence have become two of the most important resources for organizations, supporting decision-making, planning, performance evaluation, and business improvement. Data analytics is the process of examining, organizing, and interpreting data to identify useful information, patterns, and insights, while artificial intelligence extends this by enabling systems to learn from data and apply that learning to real-world tasks such as understanding customer feedback.

Through the Cisco Networking Academy, I completed three connected learning programs. Data Analytics Essentials introduced the fundamental concepts of working with data. Introduction to Modern AI built on this by explaining how artificial intelligence systems are designed, trained, and applied. Apply AI: Analyze Customer Reviews then provided a practical, applied perspective by focusing on how AI techniques can be used to analyze customer feedback and extract meaningful insights for businesses.

Together, these programs helped me build a foundation that connects data handling, analytical thinking, and applied AI — skills that are highly relevant to modern fields such as machine learning, business intelligence, software development, and information technology.

2.2 Organization / Program Profile

Cisco Networking Academy is an educational initiative associated with Cisco that provides technology-focused learning opportunities aimed at developing technical knowledge and skills relevant to the digital economy. It offers structured educational content across areas such as networking, cybersecurity, programming, data analytics, and artificial intelligence.

Data Analytics Essentials was offered by IILM University through the Cisco Networking Academy program, under the guidance of Instructor Ms. Swati Vashisht. Introduction to Modern AI and Apply AI: Analyze Customer Reviews were offered directly through the Cisco Networking Academy, under Director Lynn Bloomer. All three certificates confirm successful completion of the respective courses by the student, Gopal Agarwal.

2.3 Problem Statement

Modern organizations generate large quantities of data through their daily activities and digital systems, including growing volumes of unstructured data such as customer reviews and feedback. Raw data by itself does not automatically provide useful information — the challenge lies in organizing, examining, interpreting, and, increasingly, applying AI techniques to convert this data into meaningful insights.

Students and professionals therefore need both a foundational understanding of data analytics and a working knowledge of how AI can be applied to real business problems, such as understanding what customers are saying about a product or service. Without this combined knowledge, it is difficult to identify patterns, evaluate large volumes of feedback efficiently, and support data-driven decisions.

The purpose of this learning experience was to develop fundamental knowledge of data analytics, gain an understanding of modern AI concepts, and apply that knowledge practically to a real-world use case — analyzing customer reviews.

2.4 Project / Training Objectives

- To understand the fundamentals of data analytics and its role in modern organizations.
- To develop analytical and data-oriented thinking.
- To understand core concepts of modern artificial intelligence, including how AI systems learn from and process data.
- To learn how AI can be applied practically to analyze unstructured data such as customer reviews.
- To understand how analytical and AI-driven insights can support business decision-making.
- To develop a foundation for advanced studies in data science, machine learning, and applied AI.
- To connect theoretical knowledge with practical, applied AI use cases.
- To improve technical and problem-solving skills related to data and AI.

2.5 Scope of the Program

The scope of this combined learning experience extends across both foundational and applied domains. The concepts learned in Data Analytics Essentials and Introduction to Modern AI provide a theoretical foundation, while Apply AI: Analyze Customer Reviews demonstrates how these concepts translate into a real, practical application.

The knowledge gained can provide a foundation for further learning in Data Science, Machine Learning, Artificial Intelligence, Natural Language Processing, Business Intelligence, Data Visualization, and Database Management. It also helps students understand how AI-driven analysis of customer feedback can support marketing, product development, and customer experience strategies in real organizations.

2.6 Technologies and Concepts Covered

- Data analytics concepts and data interpretation
- Data handling and analytical thinking
- Fundamentals of modern artificial intelligence
- AI-based analysis of unstructured text data (customer reviews)
- Sentiment and feedback analysis concepts
- Data-driven and AI-assisted decision-making
- Cisco Networking Academy learning platform

2.7 Learning Workflow

The overall learning workflow followed across the three programs can be represented as:

Foundational Concepts (Data Analytics Essentials) → Understanding AI Systems (Introduction to Modern AI) → Applying AI to Real Data (Apply AI: Analyze Customer Reviews) → Extraction of Insights from Customer Feedback → Data-Driven and AI-Assisted Decision Making

Figure 2.1 – Data Analytics and Applied AI Learning Workflow

2.8 Methodology

Step 1 – Conceptual Learning: Understanding fundamental concepts related to data analytics and the role of data in modern technological environments, through Data Analytics Essentials.

Step 2 – Understanding AI: Learning the core principles of how modern AI systems are built and how they process information, through Introduction to Modern AI.

Step 3 – Applied Learning: Applying AI concepts practically to analyze real customer review data, through Apply AI: Analyze Customer Reviews.

Step 4 – Interpretation: Interpreting the patterns and insights identified from customer feedback rather than simply collecting data.

Step 5 – Application: Using the concepts learned as a foundation for future data analytics, AI, and machine learning projects.

2.9 Expected Outcomes

- Improved understanding of fundamental data analytics concepts.
- Working knowledge of how modern AI systems function.
- Practical understanding of applying AI to analyze customer feedback.
- Development of analytical and data-driven thinking skills.
- Stronger foundation for learning advanced data science and AI concepts.
- Better preparation for academic and industry-oriented technical projects.
- Improved understanding of how AI-driven insights support organizational decisions.

3. LEARNING OUTCOMES

Through the Data Analytics Essentials program, I developed an introductory understanding of data analytics and the role of data in technology-oriented decision-making, including awareness of the importance of data organization, interpretation, and analytical thinking.

Through Introduction to Modern AI, I gained an understanding of the core principles behind modern artificial intelligence systems, including how they are trained and applied to solve real-world problems.

Through Apply AI: Analyze Customer Reviews, I gained hands-on exposure to applying AI techniques to a practical business scenario — analyzing customer reviews to extract meaningful insights, understand sentiment, and identify patterns in feedback.

Collectively, these three programs helped me understand that data analytics and AI are not limited to collecting or processing information, but involve examining available information to derive meaningful, actionable insights. This understanding provides a strong foundation for future learning in data science, artificial intelligence, machine learning, and related areas, and has strengthened my interest in working with data and AI-driven technologies.

4. CONCLUSION

The Data Analytics Essentials, Introduction to Modern AI, and Apply AI: Analyze Customer Reviews programs, completed through the Cisco Networking Academy, together formed a valuable and connected learning experience. They helped me build a progressive understanding — starting with foundational data analytics concepts, moving to core AI principles, and culminating in a practical application of AI to real customer review data.

The knowledge gained through these programs can serve as a strong foundation for further technical learning in data science, artificial intelligence, machine learning, and business analytics. It also helped me recognize the importance of developing both theoretical understanding and practical, applied skills when working with data and AI.

Overall, successful completion of all three programs has enhanced my technical awareness and provided me with a stronger foundation for undertaking future academic projects and professional learning opportunities in data and AI-related fields.

5. BIBLIOGRAPHY / REFERENCES

- Cisco Networking Academy, Data Analytics Essentials, course learning material.
- Cisco Networking Academy, Introduction to Modern AI, course learning material.
- Cisco Networking Academy, Apply AI: Analyze Customer Reviews, course learning material.
- IILM University, Cisco Networking Academy Program.
- Certificate of Completion – Data Analytics Essentials, issued to Gopal Agarwal, completion date: 14 June 2026.
- Certificate of Completion – Introduction to Modern AI, issued to Gopal Agarwal, completion date: 12 June 2026.
- Certificate of Completion – Apply AI: Analyze Customer Reviews, issued to Gopal Agarwal, completion date: 12 June 2026.